

Shiqi (Stella) Chen

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Personal Profile

System and software engineer specializing in backend systems and platform infrastructure for healthcare applications serving 2M+ patients across 900+ hospitals and clinics. Broad technical foundation spanning distributed systems, machine learning, cryptography and security.

Experience

PocketHealth	Toronto, Canada
Internship - Software Engineer (PEY)	May 2024 - Aug 2025
- Built and maintained Kubernetes-based infrastructure on Azure, improving deployment automation and supporting containerized services, ingress controllers, and internal platform operations. - Implemented JWT authentication and gateway authorization for secure service-to-service communication in a distributed system. - Implemented Prometheus metrics and internal alerting to monitor service health and traffic across backend systems. - Designed and optimized dynamic GitLab CI/CD pipelines to modularize complex deployment workflows and significantly reduce execution time across services.	
Kua.ai	Zhuhai, China
Internship - Software Engineer	June 2023 - Aug 2023
- Reverse-engineered Midjourney's interface to build a seamless integration for Kua.ai, bridging the gap between raw generative AI capabilities and the high precision requirements for cross-border e-commerce product listing. - Collaborated across the development team to overcome technical constraints and deliver a reliable end-user experience.	
University of Toronto	Toronto, Canada
Work Study - Research Assistant (Supervisor: Prof. Qun Wang)	Jan 2025 - Apr 2025
- Built a model for the N-Vortex simulator to support simulation-based analysis and experimentation. - Validated and refined the implementation through testing, debugging, and iterative improvement.	
Work Study - Lecture Notes Assistant (Supervisor: Prof. Lisa Jeffrey)	Jan 2024 - Apr 2024
Prepared and refined lecture notes in elementary Lie theory.	
Teaching Assistant - MAT135, MAT233, MAT102, MAT224, MAT223, MAT136, CSC236, CSC263, CSC347, CSC427	Jan 2023 - Present
Led tutorials for math and CS courses. Coordinated tutorial logistics and developed course assessments to support course delivery.	

Education

University of Toronto	Toronto, Canada
Honours Bachelor of Science (CGPA: 3.95/4.00)	Sept 2021 - April 2026
- Computer Science Specialist; Information Security Specialist; Mathematical Science Minor 2022 Summer - Dean's List Scholar	
International Department, The Affiliated High School of South China Normal University	Guangzhou, China
High school	Aug 2018 - June 2021
President of Rubik's Club	
Sun Yat-sen University Affiliated Middle School	Guangzhou, China
Middle School	Sept 2015 - July 2018

Selected Coursework

MAT302 - Introduction to Algebraic Cryptography	Jan 2026 - Apr 2026
Studied algebraic and number-theoretic cryptography, focusing on public-key constructions and digital signatures, including Diffie-Hellman, ElGamal, RSA, and DSA/ECDSA.	
CSC2125 - Topics in Software Engineering: Blockchain Technology & Engineering	Sept 2025 - Dec 2025
- Modeled PBFT in TLA+ and used formal verification to analyze safety and fault tolerance in distributed consensus. - Examined smart contract semantics and their dependence on protocol-level guarantees for secure decentralized execution.	
CSC427 - Computer Security	Jan 2024 - Apr 2024
- Investigated VM detection and anti-detection techniques in the broader context of virtualization security. - Analyzed CVE-2015-5165 & CVE-2015-7504 as case studies in QEMU-based VM escape, using Debian as the guest environment.	

Skills

Languages	Go, Python, Java, Kotlin, C/C++, SQL, TypeScript, UNIX/Shell, HTML/CSS, Solidity
Infrastructure & Tools	Skills, Git, Docker, Kubernetes, Terraform, GitLab CI, Prometheus, Microsoft Azure, LaTeX
Areas	Distributed Systems, Cryptography, Blockchain, Machine Learning, Computer Security, Infrastructure Automation